



Software Process Improvement –SEPI-471



PROCESS MODELS FOR SPI

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Agenda

- Process Models for SPI
 - Capability Maturity Model Integration (CMMI)
 - ISO9001 Standard
 - ISO15504
 - PSP and TSP
 - SixSigma

Capability Maturity Model Integration (CMMI)



CMMI was developed by the Software Engineering Institute and is the successor to the older software CMM which was released in the early 1990s.

The CMMI is a suite of products used for improving processes, and it includes models, appraisal methods, and training material.

The CMMI models address three areas of interest

CMMI for Development (CMMI-DEV)- *for software development organizations*

CMMI for Services (CMMI-SVC)- *for service providing organizations*

CMMI for Acquisition (CMMI-ACQ)-*acquisition of desired skills*

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Capability Maturity Model Integration (CMMI)



The CMMI Development Model is the vehicle used for software process improvement

It is a framework that **allows organizations to improve their maturity by improvements to their underlying processes.**

It provides a **structured approach and allows the organization to set improvement goals and priorities.**

It provides a clearly defined roadmap for improvement and it allows the organization to improve at its own pace.

It allows the processes to evolve from ad hoc immature activities to **disciplined mature processes**

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ISO9001

ISO9001 is an internationally recognized **quality management standard** and is customer and process focused.

It applies to the processes that an organization uses to create and control products and services, and it emphasizes continuous improvement.

The standard is designed to apply to any product or service that an organization supplies.

The implementation of ISO 9001 involves understanding the requirements of the standard and how the standard applies to the organization

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ISO9001

It requires the organization to identify its quality objectives, define a quality policy, produce documented procedures, and carry out independent audits to ensure that the processes and procedures are followed.

An organization may be certified against the ISO 9001 standard to gain recognition to its commitment to quality and continuous improvement.

ISO9001

The certification involves an independent assessment of the organization to verify that it has implemented the ISO 9001 requirements properly and that the quality management system is effective.

It will also verify that the processes and procedures defined are consistently followed and that appropriate records are maintained.

The ISO 9004 standard provides guidance for continuous improvement.

ISO/IEC 15504

The ISO/IEC 15504 standard (popularly known as ISO SPICE)(Software Process Improvement and Capability Determination) is an international standard for process assessment.

It includes guidance for process improvement and for process capability determination, as well as guidance for performing an assessment.

It includes an exemplar process assessment model for software life cycle processes and also an exemplar process assessment model for systems life cycle processes.

There is a version of SPICE termed “Automotive SPICE” that is used in the automotive sector.

ISO/IEC 15504 can be used in a similar way to the CMMI and its exemplar models (for either software or systems life cycles) may be employed to implement best practice in process definition.

Assessments may be performed to identify strengths and opportunities for improvement.

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Personal Software Process (PSP)

The Personal Software Process (PSP) is a disciplined data-driven software development process that is designed to help software engineers understand and to improve their personal software process performance.

It was developed by Watt Humphrey at the SEI, and it helps engineers to improve their estimation and planning skills and to reduce the number of defects in their work.

This enables them to make commitments that they can keep and to manage the quality of their projects.

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Team Software Process (TSP)

The Team Software Process (TSP) was developed by Watt Humphries at the SEI.

It is a structured approach designed to help software teams understand and improve their quality and productivity.

Its focus is on **building an effective software development team, and it involves establishing team goals, assigning team roles as well as other teamwork activities.**

Team members must already be familiar with the PSP.

Six Sigma (6σ)

Six Sigma (6σ) was developed by Motorola as a way to improve quality and reduce waste.

Its approach is to identify and **remove the causes of defects in processes by minimizing process variability.**

Six Sigma (6σ)

It uses quality management techniques and tools such as the

-5 Whys

-Business process mapping

-DMAIC and DMADV methodologies

- Define, Measure, Analyze, Improve and Control -DMAIC
- Define, Measure, Analyze, Design, and Verify.
- DMAIC reduces/eliminates defects (reactive); DMADV prevents defects (proactive)

Six Sigma (6σ)

There are several roles involved in six-sigma initiatives such as

- Champions
- Black Belts
- Green Belts
- Yellow Belt
- White Belt

Each role requires knowledge and experience and is awarded on merit subject to training and certification.

Six Sigma was influenced by earlier quality management techniques developed by Shewhart, Deming, and Juran.



Thank you

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